

Mayfew Features + Labels

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From EnergyStats Files

- Sniper Totals (add up across energy-stats files)
 - cdb alu accesses
 - dcache read accesses
 - total real instructions (total ins - NOPs)
 - ROB reads
 - Busy cycles
 - icache read accesses
 - committed int instructions
 - dtlb total accesses
 - itlb total misse
 - BTB read accesses
 - dcache read misses
 - cdb fpu accesses
 - branch mispredictions
 - dcache write accesses
- All of the previous totals divided by number of cycles in the interval added from all of the energystats files. This gives us a Real IPC number
- Avg MUL cdb duty cycle (add up duty cycle across energystats files, divide by number of energystats files)
- Avg LSU duty cycle
- Avg IFU duty cycle
- Avg FPU cdb duty cycle
- Peak Duty cycle for each of the duty cycles divided by avg of each of the duty cycles (should get one peak/average for each duty cycle)
- Max real IPC of the energystats files divided by real IPC of all of them

From BBV files

- Dependency chain features
 - Build a DAG for each BB inside of the interval. Each instruction is a node and there is a directed edge if one instruction relies on another
 - Get longest dependency chain for each BB, call it CP_{BB} .
 - Normalize it by dividing by the number of instructions N_{BB} in the BB to get $NormCP_{BB}$.
 - get ILP proxy ILP by doing N_{BB}/CP_{BB}
 - Let f_b be the number of times the BB executes in the interval, N_b be number of ins in BB b , and CP_b be the longest critical path in BB b . Then $DynInst_b = f_b(N_b)$.
 - Calculate three features per interval from this setup
 1. $MeanCP_i = \frac{\sum DynInst_b(CP_b)}{\sum DynInst_b}$
 2. $MeanNormCP_i = \frac{\sum DynInst_b(CP_b/N_b)}{\sum DynInst_b}$
 3. $MeanILP_i = \frac{\sum DynInst_b(N_b/CP_b)}{\sum DynInst_b}$
 - Max critical path for the interval
 - percentage of instructions in critical paths of 3 or longer. Calculate instructions from the BBV files, not the snipersim files
- Instruction-level entropy/diversity features
 - Entropy $H = -\sum p_b \log p_b$ summing over basic blocks
 - $p_b = \frac{DynInst_b}{\sum DynInst}$
 - Do this for instruction types found in the Basic Blocks
 - * $p_c = \frac{DynInst_c}{\sum DynInst_k}$ where c refers to a category of instruction.
 - * These are the instruction types that I want an entropy calculation for
 1. Integer ALU
 2. MUL/DIV
 3. FP
 4. SIMD
 5. LOAD
 6. STORE
 7. BRANCH
 - * It is important to normalize these by number of categories
 - Percentage of instructions that are serializing instructions. Serializing instructions match these OPCODES:

1. cpuid
2. mfence
3. lfence
4. sfence
5. rdtsc
6. wrmsr
7. call
8. ret

Intent vs Execution

- For ALU, FPU, LSU, find from BBVs percentage of dynamic instructions that belong to ALU, FPU, LSU
- compare to Sniper duty cycle for an interval:
 - $Mismatch = \frac{\text{ALU duty cycle}}{\text{fraction of BB ints are ALU ins}}$
 - Do the same for LSU and FPU

Labels

- Peak temperature (find greatest value from die_grid.temps)
- Average tempeature (average value from die_grid.temps)
- Max positive MLTD from die_grid.temps.2dmaxima.csv
- Hotspot severity score:

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$$\sigma(x; x_0, y_0, s, a) = \frac{a}{1 + e^{-s(x-x_0)}} + y_0$$

$$\sigma_{df}(T) = \sigma(T; 115, 0, 0.2, 2)$$

$$\sigma_M(\text{MLTD}) = \sigma(\text{MLTD}; 25, -0.25, 0.2, 1.25)$$

$$\sigma_T(T) = \sigma(T; 60, 0.35, 0.05, 0.65)$$

$$\text{sev}(x, y) = \sigma_{df}(T_{x,y}) + \sigma_M(\text{MLTD}_{x,y}) \cdot \sigma_T(T_{x,y})$$